Foundations of Artificial Intelligence - 2026.1

Sequential Decisions under Uncertainty - Introduction to Reinforcement Learning

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Decision Making Situations

Decision Situation Matrix

Action Effects

Horizon		**Certainty**	**Risk**	**Uncertainty**
	Atomic	✓	✓	✓
	Sequential	✓	✓	NOW

Class Overview

1. Requirements and the GPI Algorithm for Unknown MDPs
2. Policy Evaluation Options for RL GPI
3. Resolving Technical Issues

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- he knows
  - the *current state* he is in and
  - the *actions he may take* (but not necessarily the successor states let alone the transition probabilities).
- a trial interface
  1. tells him about possible actions,
  2. receives his choices, and
  3. informs him about the successor state and the observed reward.

Reinforcement Learning

Why Standard PE does not work anymore

For a known MDP, we can do PE solving Bellman or using it as an update rule:

$$v^\pi(s) = r(s) + \gamma \sum_{s'} \mathbb{P}(s'|s, \pi(s)) v^\pi(s')$$

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But these also involve the probabilities $\mathbb{P}$ (and maybe rewards r).

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Open question: How can we use that data to update estimates $\hat{q}(s, a)$?

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In practice, we will not pad, this is just to make sure that τ is conceptually defined up to time T for any generated trial.

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For this trial τ we can compute an *observed utility*

$$\hat{U}_\tau = \sum_{t=0}^{T} \gamma^t r(s_t)$$

$\hat{U}_\tau$ is a *sample* from the RV $r(s_0) + \gamma U\left((S_t)_{t\in\mathbb{N}}^{S_1,\pi}\right)$, where $S_1 \sim \mathbb{P}(\cdot|s_0, a_0)$.

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But $q(s_0, a_0) = \mathbb{E}^{s_0,a_0}[r(S_0) + \gamma U\left((S_t)_{t\in\mathbb{N}}^{S_1,\pi}\right)]$ is the expected value of this RV.

Policy Evaluation in Unknown MDPs

Creating and Aggregating Trials

Recall: for any random variable X for which we have i.i.d. samples $\hat{X}_1, .., \hat{X}_n$, the empirical mean is an *unbiased*, *consistent*, and *quickly concentrating* estimate.

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We can estimate $q^\pi(s_0, a_0) = \mathbb{E}^{s_0,a_0}[r(S_0) + \gamma U\left((S_t)_{t\in\mathbb{N}}^{S_1,\pi}\right)]$ if we

1. draw a reasonably large number of trials $\tau_1, .., \tau_n$ starting with (s_0, a_0) and
2. estimate $\hat{q}^\pi(s_0, a_0) := \frac{1}{n}\sum_{i=1}^n \hat{U}_{\tau_i}$.

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The efficient update after the n-th incoming trial τ_n starting with (s_0, a_0) would be

$$\boxed{\hat{q}(s_0, a_0) \ += (\hat{U}_{\tau_n} - \hat{q}(s_0, a_0)) \ / \ n}$$

Policy Evaluation in Unknown MDPs

Efficient Usage of Trial Data

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Any trial τ could give us a one-point estimate of $\hat{q}(s_i, a_i)$ for *each* pair (s_i, a_i) in τ via

$$\boxed{\hat{U}_{\tau[i:]} := \sum_{t=i}^{T} \gamma^{t-i} r(s_t)}$$

where $\tau[i:]$ is τ ignoring the first i time steps.

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So we can use τ to update not 1 but up to $|\tau|$ values $\hat{q}(s_i, a_i)$.

Policy Evaluation in Unknown MDPs

Direct Utility Estimation - Idea

If we apply the logic for each state, then a state-action pair might occur several times in the same trial.

In principle, we can update the estimate $\hat{q}(s, a)$

1. on *each* occurrence of (s, a) in the trial (Every-Visit Monte Carlo (EVMC))
2. only for the *first* occurrence of (s, a) in the trial (First-Visit Monte Carlo (FVMC))

Policy Evaluation in Unknown MDPs

First-Visit Monte Carlo for Fixed Initial State and Fixed Policy

Require: policy π, trial_gen, s_0, T, $\hat{q}$

Ensure: update of $\hat{q}$ that will converge towards q^π if invoked an infinite number of times in sequence for π

```
N = ∅ // this might also be a memorized resource across multiple PE steps
repeat
    Generate a trial τ = s_0, a_0, r_0, .., s_|τ|, a_|τ|, r_|τ| using trial_gen with π for at most T steps
    U ← r_|τ|
    for t = |τ| − 1, .., 0 do
        U ← γU + r_t
        if s_t, a_t does not appear in s_0, a_0, s_1, a_1, .., s_{t−1}, a_{t−1} then
            if (s_t, a_t) ∉ N then
                Initialize q̂(s_t, a_t) and N[(s_t, a_t)]
            end if
            N[(s_t, a_t)] += 1
            q̂(s_t, a_t) += (U − q̂(s_t, a_t)) / N[(s_t, a_t)]
        end if
    end for
until PE step should end
```

Policy Evaluation in Unknown MDPs

FVMC - Discussion

Advantages:

- Very fast due to asynchronous state updates.
- No model needs to be learned, so also applicable in large MDPs.

Disadvantages:

- Possibly inefficient since Bellman equation is not being used.
- No knowledge about the MDP (beyond q-values) is being generated.

Policy Evaluation in Unknown MDPs

Adaptive Dynamic Programming (ADP)

One approach to update $\hat{q}(s, a)$ is to *learn* $\mathbb{P}$ and r from trials.

With estimates $\hat{\mathbb{P}}$ and r, we update $\hat{v}$ and next $\hat{q}$ via standard PE.

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Learning $\mathbb{P}$ and r is conceptually easy if we keep track of the number of times $N[s][a][s']$ we were transported from s to s' using action a.

This is form of policy evaluation is called Adaptive Dynamic Programming (ADP).

Policy Evaluation in Unknown MDPs

Adaptive Dynamic Programming (ADP)

Require: policy π, trial_gen, s_0, T, $\hat{q}$

Ensure: update of $\hat{q}$ that will converge towards q^π if invoked an infinite number of times in sequence for π

$S = []$, $N = \emptyset$, $rewards = []$ // this might also be a memorized resource across multiple PE steps
repeat
    Generate a trial $\tau = s_0, a_0, r_0, .., s_{|\tau|}, a_{|\tau|}, r_{|\tau|}$ using trial_gen with π for at most T steps
    for $t = 1, .., T$ **do**
        if $s_t \notin S$ **then**
            S.append(s_t)
            expand N with a row and column of 0s
            rewards.append(r_t)
        end if
        $N[s_{t-1}][a_{t-1}][s_t] + +$
        $\hat{\mathbb{P}}(s_t|s_{t-1}, a_{t-1}) := N[s_{t-1}][a_{t-1}][s_t] / \sum_{s'} N[s_{t-1}][a_{t-1}][s']$
    end for
    $\hat{v}^\pi \leftarrow$ MDP-Policy-Evaluation($\widehat{\mathbb{P}}$, $rewards$, π) // internal update, π has no access
until no more PE steps
update $\hat{q}$ from $\hat{v}^\pi$, $\hat{\mathbb{P}}$, and $rewards$

Policy Evaluation in Unknown MDPs

Adaptive Dynamic Programming (ADP) - Discussion

Advantages:

- Learns the MDP properties itself, which can be useful if the policy is changed (because the MDP properties are independent of π and can be reused for estimates of other policies later on).
- Knowledge can be transferred.

Disadvantages:

- might require a lot of memory
- updating the probabilities can become slow

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Suppose that all trials start in s_0 and follow π all the time.

Then we will be unable to learn *anything* for $q^\pi(s, a)$ if

- $a \neq \pi(s)$ or
- s is not reachable from s_0 via π.

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Then we will be unable to learn *anything* for $q^\pi(s, a)$ if

- $a \neq \pi(s)$ or
- s is not reachable from s_0 via π.

So it is clear that we cannot create trials just be following π from s_0.

With this we could only estimate v^π, but this is not enough without $\mathbb{P}$.

Policy Evaluation in Unknown MDPs

Exploring Starts

Maybe obvious solution: pick random actions occasionally instead of following π.

The challenge with this is that it defines an alternative policy π'.

So trials would give us data to estimate $q^{\pi'}$ instead of q^{π}.

If π' is similar to π or converges to π^* in the limit, this might be ok.

But maybe we can do something easier first.

Policy Evaluation in Unknown MDPs

Exploring Starts

Since $q^\pi(s, a)$ is the expected utility of starting in s, using action a *once* and then following π, it would be reasonable to

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This technique is called Exploring Starts.

In the limit, it tries each action $a \in A_s$ infinitely often in every state $s \in S$.

Avoiding Infinite Trials

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- what are the implications of doing that?
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- what could be T?

The answer to the first question is simple: We pretend that $r(S_t) = 0$ for all $t > T$.

Of course this is not necessarily so in the true MDP (and we usually cannot know).

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$$\boxed{T \geq \log_\gamma \frac{\varepsilon(1-\gamma)}{\max|r|} \quad \Rightarrow \quad \left|\sum_{t=T}^{\infty} \gamma^t r(S_t)\right| < \varepsilon}$$

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where $\max|r|$ is the biggest absolute value of any reward in the MDP.

This guarantees that

$$|\hat{U}_\tau - \hat{U}_{\tau[:T]}| < \varepsilon$$

So we can make sure that we approximate the true utilities of the trials up to an arbitrary precision ε while being allowed to stop trial generation after T steps.

Avoiding Infinite Trials

Based on these observations, we will always make the following assumption:

> *There is some known $T \in \mathbb{N}$ so that the quality of all trials $\{\tau \mid \tau[: T] = \tau_0\}$ with the same T-prefix τ_0 are considered equally as good as τ_0.*

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This is for example the case in many

1. games that will end after at most a certain number of turns or time steps, and
2. situations where $\gamma < 1$ with
   - known upper bound for the rewards and
   - rewards distributed across trials (not just in end states).

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This is specifically problematic if all rewards are 0 except in a final state, which is maybe never reached.

Since we *must* define an upper limit for trial lengths, we might want to introduce synthetic rewards to guide the process; this is called Reward Shaping.

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Summarizing the class in one sentence,

- standard PE is not feasible in situations under uncertainty, because we don't know the transition probabilities and rewards;
- we can address this by directly estimating q-values from trials (FVMC) or learn the MDP properties from trials,
- but we must make sure that trials end after a finite number of steps and also that each action is sometimes tried in each state, regardless of π.